Introduction to Deep Learning

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What is Deep Learning?

Definition

Deep Learning is a subfield of Machine Learning that uses multi-layer artificial neural networks to automatically learn representations from data.

- Learns hierarchical features directly from raw data.
- Uses multiple hidden layers to represent complex patterns.
- Inspired by the structure and functioning of the human brain.

Input Data $\rightarrow$ Feature Learning $\rightarrow$ Prediction

Machine Learning vs Deep Learning

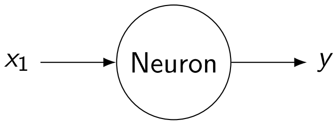
Traditional Machine Learning

- Requires manual feature extraction.
- Works well with small and structured datasets.
- Examples:
  - SVM
  - Decision Tree
  - Logistic Regression

Deep Learning

- Automatically learns features.
- Requires large datasets.
- Uses deep neural networks.
- Examples:
  - CNN
  - RNN
  - Transformer

From Neuron to Artificial Neural Network



An artificial neuron performs:

$$z = \sum_{i=1}^{n} w_i x_i + b$$

$$y = f(z)$$

where:

- w_i = weights
- b = bias
- $f(.)$ = activation function

Shallow Network vs Deep Network

Shallow Neural Network

$$Input \rightarrow Hidden \rightarrow Output$$

- One hidden layer
- Limited feature learning

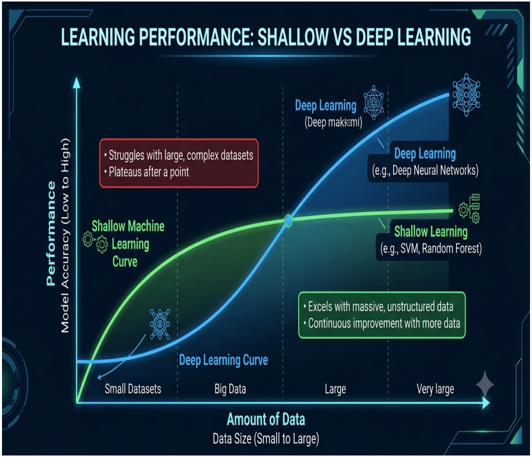
Deep Neural Network

$$Input \rightarrow H_1 \rightarrow H_2 \rightarrow H_3 \rightarrow Output$$

- Multiple hidden layers
- Learns hierarchical representations

Learning Capability: Shallow vs Deep Networks

- Both models improve as more data becomes available.
- **Shallow networks** learn quickly but eventually reach a performance limit.
- **Deep networks** continue improving by learning complex and hierarchical features.
- Deep learning usually requires larger datasets to outperform shallow models.



How Does Deep Learning Work?

1. **Forward Propagation**

   $$Input \rightarrow HiddenLayers \rightarrow Output$$

   Network produces prediction.
2. **Loss Calculation**

   $$Loss = Difference(Target, Prediction)$$
3. **Backpropagation**
   Gradients are calculated to update weights.
4. **Optimization**
   Algorithms such as Gradient Descent and Adam minimize the loss.

Major Deep Learning Architectures

Architecture	**Best For**	**Examples**
MLP	Tabular Data	Classification, Regression
CNN	Images	Object Detection, Medical Imagi
RNN/LSTM	Sequential Data	Speech, Time Series
Transformer	Long Sequences	NLP, LLMs, Vision
GAN	Data Generation	Image Synthesis
Autoencoder	Representation Learning	Compression, Denoising

Multilayer Perceptron (MLP)

Definition

A Multilayer Perceptron (MLP) is a feedforward neural network consisting of an input layer, one or more hidden layers, and an output layer.

$$\text{Input} \rightarrow H_1 \rightarrow H_2 \rightarrow H_3 \rightarrow \text{Output}$$

- Fully connected architecture.
- Uses activation functions (ReLU, Sigmoid, Tanh).
- Suitable for classification and regression tasks.
- An MLP with multiple hidden layers is called a **Deep MLP**.

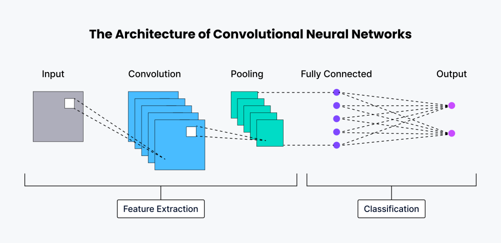
Convolutional Neural Network (CNN)

What is a CNN?

- A deep learning model designed for image and video data.
- Automatically learns spatial features from images.
- Uses convolutional filters to detect patterns such as edges, textures, and shapes.

Main Layers

- Convolution Layer
- Pooling Layer
- Fully Connected Layer



How CNN Processes an Image

Working of a CNN

1. **Input Layer:** Receives the input image.
2. **Convolution Layer:** Applies filters to extract features such as edges and textures.
3. **Activation (ReLU):** Introduces non-linearity into the model.
4. **Pooling Layer:** Reduces the feature-map size while preserving important information.
5. **Fully Connected Layer:** Combines extracted features to predict the output class.

Applications

- Image classification
- Face recognition
- Medical image analysis
- Object detection
- Autonomous vehicles

Recurrent Neural Network (RNN)

Unlike MLPs, RNNs remember previous information.

$$x_1 \to x_2 \to x_3 \to x_4$$

Each prediction depends on previous inputs.
Applications:

- Speech recognition
- Language modeling
- Time-series forecasting

Limitation:

- Difficult to learn long-term dependencies.

Long Short-Term Memory (LSTM)

LSTM is an improved version of RNN.
Advantages:

- Stores long-term information
- Reduces vanishing gradient problem
- Uses memory cells and gates

Applications:

- Machine translation
- Stock prediction
- Speech recognition

Transformer

Unlike RNNs, Transformers process all words simultaneously.
Key idea:

$$\boxed{\text{Self-Attention}}$$

Advantages

- Parallel computation
- Captures long-range dependencies
- Faster training
- Foundation of ChatGPT, BERT and GPT

Generative Adversarial Network (GAN)

Definition

A Generative Adversarial Network (GAN) is a deep learning model that learns to generate new data samples that resemble real data by training two neural networks in competition.

Generator (G)

- Takes random noise as input.
- Generates synthetic (fake) data.
- Learns to produce realistic samples.

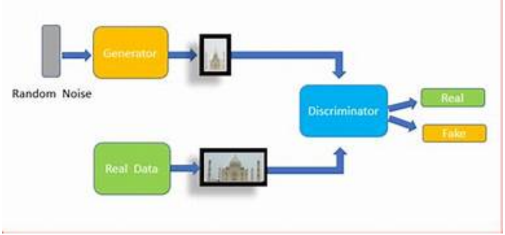
Discriminator (D)

- Receives both real and fake samples.
- Predicts whether a sample is real or fake.
- Provides feedback to improve the generator.

How Does a GAN Work?

Training Steps

1. Generate random noise.
2. Generator creates a fake image.
3. Discriminator compares real and fake images.
4. Both networks update their parameters.
5. Repeat until generated images become highly realistic.



Applications

Image generation . Face synthesis . Data augmentation . Super-resolution

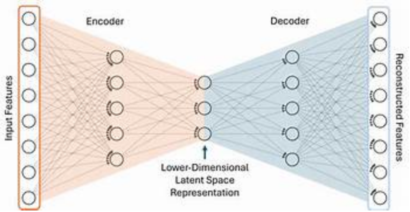
Autoencoder

Definition

An autoencoder is an unsupervised neural network that learns a compressed representation of data and reconstructs the original input.

Key Components

- **Encoder:** Compresses the input.
- **Latent Space:** Stores essential features.
- **Decoder:** Reconstructs the input.



Input $\rightarrow$ Encoder $\rightarrow$ Latent $\rightarrow$ Decoder $\rightarrow$ Output

Applications: Image denoising • Data compression • Anomaly detection • Feature extraction

Comparison of Deep Learning Architectures

Architecture	Data Type	Learning Task	Typical Applications
MLP	Tabular	Classification/Regression	Disease prediction, credit scoring, house price prediction
CNN	Images	Feature Extraction	Image classification, object detection, medical imaging
RNN/LSTM	Sequential	Sequence Modeling	Speech recognition, language modeling, time-series forecasting
Transformer	Text/Sequence	Attention-based Learning	Machine translation, chatbots, Large Language Models (LLMs)
GAN	Images, Audio	Data Generation	Image synthesis, deepfakes, super-resolution, data augmentation
Autoencoder	Any Data	Representation Learning	Denoising, compression, anomaly detection, feature extraction